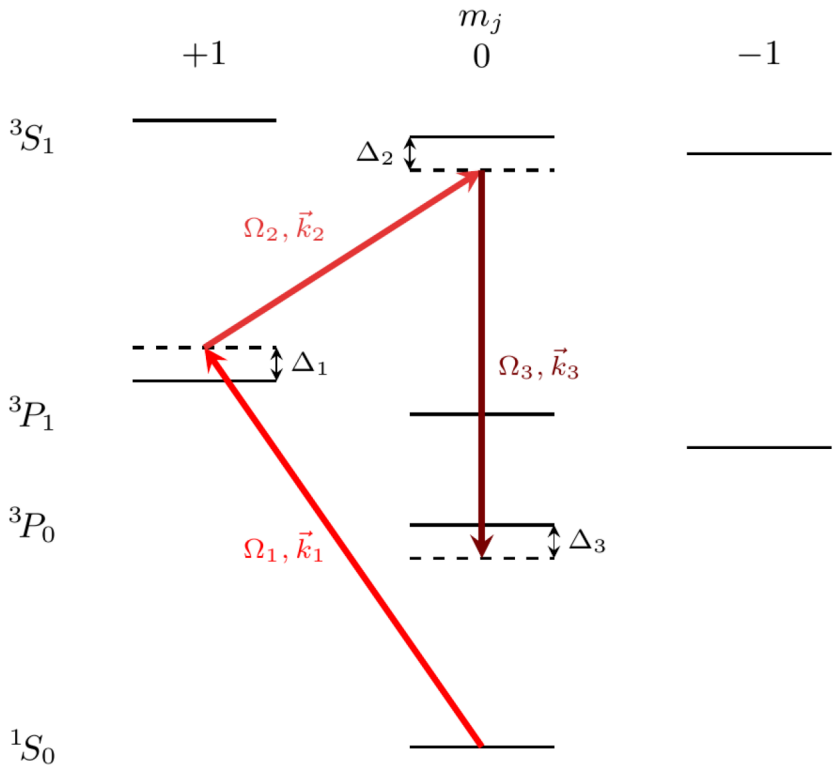
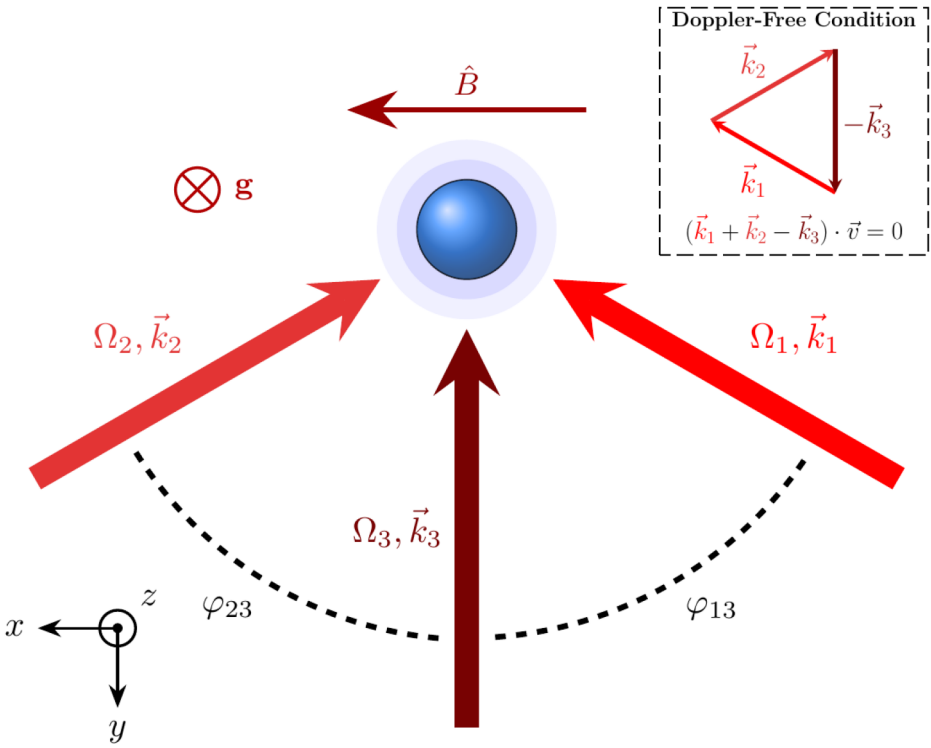
**(a)**


$m_j$
$+1$
$0$
$-1$
$^3S_1$
$\Delta_2$
$\Omega_2, \vec{k}_2$
$\Delta_1$
$\Omega_3, \vec{k}_3$
$^3P_1$
$^3P_0$
$\Omega_1, \vec{k}_1$
$\Delta_3$
$^1S_0$

**(b)**


Doppler-Free Condition
$\vec{k}_2$
$-\vec{k}_3$
$\vec{k}_1$
$(\vec{k}_1 + \vec{k}_2 - \vec{k}_3) \cdot \vec{v} = 0$
$\hat{B}$
g
$\Omega_2, \vec{k}_2$
$\Omega_1, \vec{k}_1$
$\Omega_3, \vec{k}_3$
$\varphi_{23}$
$\varphi_{13}$
$x$
$z$
$y$